# Yili Wen

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The University of Texas at Dallas
Erik Jonsson School of Engineering and Computer Science
Design & Engineering for Making (DE4M) Lab

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My research in Human-Computer Interaction (HCI) focuses on integrating tangible user interfaces, wearable technology, and artificial intelligence into everyday human experiences. I develop interactive systems that bridge physical and digital interactions. Technically, I specialize in prototyping hardware-software systems, sensor integration, and AI applications for interactive experiences. My methodology is versatile: I employ mixed-methods research, combining qualitative approaches (interviews, observational studies) with quantitative techniques (sensor data analysis, surveys, experimental evaluation) through lab and field studies to design, implement, and critically assess real-world prototypes.

## EDUCATION

8/2030 (Expected) **Ph.D. in Computer and Information Science,** *The University of Texas at Dallas, Richardson, TX, USA*
Advisor: Liang He

5/2025 **B.A. in Computation and Design,** *Duke Kunshan University, Kunshan, China*
**B.A. in Interdisciplinary Studies,** *Duke University, Dual Degree Undergraduate Program*
Advisor: Xin Tong

**Awards**

2023 Center for the Study of Contemporary China (CSCC) Fellowship

N/A Dean's List - Duke Kunshan University (Fall 2022, Fall 2023, Spring 2024, Fall 2024)

## RESEARCH

### Publications

***Conference Papers***

2025 [C.1] "I Want to Use Technology to Help Farmers in the Future": Enhancing Children's Understanding of Farming through Tangible User Interfaces and Embodied Interaction. **Yili Wen**, Jiaxin Wang, Hongni Ye, Liang He, Min Fan, Xin Tong. In Proceedings of the 24th Annual ACM Interaction Design and Children Conference - **IDC'25**.

2024 [C.2] "I Keep Sweet Cats In Real Life, But What I Need In The Virtual World Is A Neurotic Dragon": Virtual Pet Design With Personality Patterns. Hongni Ye, Ruoxin You, Kaiyuan Lou, **Yili Wen**, Xin Yi, Xin Tong. In Proceedings of the 11th Annual ACM International Symposium of Chinese CHI - **Chinese CHI' 24**.

***Extended Abstracts/Poster& Workshop Papers***

2026 [E.1] PufFab: Making Shape Transformable Rice Paper for Playful Food Fabrication. **Yili Wen**, Yafei Ge, Xin Tong, and Liang He. To appear at - **TEI '26**.

2023 [W.1] PetGen: Design and Generation of Virtual Pets. Hongni Ye, Ruoxin You, Kaiyuan Lou, **Yili Wen**, Xin Yi, Xin Tong. In 2023 IEEE International Conference on Multimedia and Expo Workshops - **ICMEW'23**.

### Research Experience

8/2025-Present **Research Assistant** *with Liang He*
*Design & Engineering for Making (DE4M) Lab at the University of Texas at Dallas, TX, USA*

1/2023-7/2025 **Research Assistant** *with Xin Tong*
*Information Hub at the Hong Kong University of Science and Technology (Guangzhou), China*
*Human-Computer Interaction Lab at Duke Kunshan University, China*

10/2023-3/2025 **Research Assistant** *with Karen Cochrane*
*Embodied Computing Lab at the University of Waterloo, Canada*

10/2022-9/2023 **Research Assistant** *with Fan Liang*
*Center for the Study of Contemporary China at Duke Kunshan University, China*

## Research Projects

10/2025-Present (Ongoing)

**From Intention to Action: A Brain-Controlled Exoskeleton Pipeline for Relearning Functional Hand Skills**

- Developing a brain signal simulator based on current literature to model neural intentions for motor control.
- Designing and fabricating a hand exoskeleton specifically for Spinal Cord Injury (SCI) patients to assist with Activities of Daily Living (ADLs).
- Establishing a pipeline to bridge high-fidelity neural decoding with adaptive wearable assistance.

7/2025-12/2025 [E.1]

**PufFab: Making Shape Transformable Rice Paper for Playful Food Fabrication**

- Designed an interactive design-to-fabrication pipeline utilizing Vietnamese rice paper for customizable edible shape transformation.
- Modified 3D printer hardware to enable the fabrication of 2D patterns that transform into complex 3D structures through frying.

1/2025-8/2025 (In Submission)

**Human-AI Co-Writing in Knowledge-Intensive Tasks: Behaviors, Challenges, and Collaboration Patterns**

- Coordinated and conducted a within-subject user study with 23 participants to evaluate LLM-human collaboration patterns.
- Analyzed extensive qualitative data from writing activity logs and interviews to identify distinct collaboration behaviors.
- Synthesized final collaboration patterns that reflect how personal intentions shape the use of AI in knowledge work.

10/2023-5/2025 (In Submission)

**Comparing Narrative Inquiry Methods to Support the Design of Game Controller Adaptive Accessories for People with Upper Limb Motor Disabilities**

- Built adaptive game controller prototypes to serve as the primary hardware for user evaluation.
- Performed qualitative analysis on interview data to compare thematic and dialogic/ performance analysis methods.
- Evaluated the efficacy of narrative inquiry in capturing the lived experiences of users with upper-limb motor disabilities.

5/2023-9/2024 [C.1]

**"I Want to Use Technology to Help Farmers in the Future": Enhancing Children's Understanding of Farming through Tangible User Interfaces and Embodied Interaction**

- Spearheaded the design and development of WooGu, an educational system integrating digital devices, Tangible User Interfaces (TUIs), and embodied interactions to teach agricultural literacy.
- Architected the interaction model by bridging 3D-printed tools with integrated sensors to provide multi-modal feedback (visual, haptic, and digital), encouraging children's physical engagement with farming concepts.

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## TEACHING EXPERIENCE

8/2025-12/2025 **Teaching Assistant**

*Introduction to Human-Computer Interaction at Department of Computer Science*

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## Skills

**Development**

- Programming: Python, R, CSS, HTML, JavaScript/p5js, Java/Processing
- Applications and Hardware: XCode, Arduino, Blender, Tableau

**Design**

- Figma, Adobe Premiere, Adobe Photoshop

**Research Methodologies**

- Quantitative, Qualitative, and Mixed-method Studies (Field Study, Case Study)